EXP-9 : To decrypt the supplied PT-109 message using Playfair.

PROGRAM :

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>
int main(void) {
char box[] = "ROYALNEWZDVBCFGHIKMPQSTUX";
char c[] = "KXJEYUREBEZWEHEWRYTUHEYFS"
"KREHEGOYFIWTTTUOLKSYCAJPO"
"BOTEIZONTXBYBNTGONEYCUZWR"
"GDSONSXBOUYWRHEBAAHYUSEDQ";
for (int i = 0; c[i]; i += 2) {
char x = c[i] == 'J' ? 'I' : c[i];
char y = c[i+1] == 'J' ? 'I' : c[i+1];
if (x == 'T' && y == 'T') { printf("TT"); continue; }
int a = (int)(strchr(box, x)-box);
int b = (int)(strchr(box, y)-box);
int r = a/5, s = b/5, u = a%5, v = b%5;
if (r == s) { u = (u+4)%5; v = (v+4)%5; }
else if (u == v) { r = (r+4)%5; s = (s+4)%5; }
else { int t = u; u = v; v = t; }
printf("%c%c", box[5*r+u], box[5*s+v]);
}
puts("");
return 0;
}
```

OUTPUT :

input

```
PTBOATONEOWENINELOSTINACTIONINBLACKETTSTRAITTWOMILESSWMERESUCOVEXCREWOFTWELVEXREQUESTANYINFORMATIONX


...Program finished with exit code 0
Press ENTER to exit console.
```